Strong Baselines, Data Provenance and Post-Development Evaluation of Daily Mandi Price Forecasts at National Scale

C Rishitha, Karthikeya Y, Ashwin Tyagi and Kaushal S

***Abstract*—We evaluate daily wholesale price forecasts for 11,845 Indian mandi series (36 commodities) at four horizons. The evaluation uses target-purged chronological splits, strong baselines, a provenance audit and post-development scoring. Persistence is hard to beat, because a fifth to a third of price changes are exactly zero. Among global learners only absolute-loss models beat it at the next report, and gains reach 3–8%. A public mirror holds only 30% of official reports from October 2025, around a change of publishing portal. On the mirror's own rows, interval coverage drops to 68–72%; on the complete record the same rows give 79–80%. Frozen models still beat persistence on six later months of official reports. Retraining the Spark models on two further years gave no measurable gain. A global LightGBM beat the Spark models and, under a pre-registered rule, replaced them in service. Provenance audits should precede claims of drift.**

***Index Terms*—Agricultural prices, Forecast evaluation, Data provenance, Prospective evaluation, Conformal prediction, Benchmarks**

I. Introduction

A large share of Indian agricultural produce is traded through regulated wholesale markets (*mandis*), although much is also sold outside them. Each mandi publishes a daily minimum, maximum and modal price per commodity and variety through AGMARKNET, mirrored on the Open Government Data platform. These reports are the most detailed public price signal available to a farmer deciding whether to sell today, wait, or travel to another market.

Forecasting them has a large literature (Section II), dominated by studies of one market, a few markets, or monthly indices. These studies typically use a single proportional hold-out, report deep or hybrid models against ARIMA, and rarely report naive baselines or prediction intervals [1]–[3]. Forecast-evaluation research shows why this matters:

- simple benchmarks are hard to beat [4], [5];
- scaled errors are needed across heterogeneous series [6];
- the point forecast that is optimal depends on the scoring function [7], [8];
- leakage and data vintage can dominate conclusions [9], [10].

We ask how much of the apparent skill in mandi price forecasting survives an evaluation that is leakage-resistant, baseline-honest, provenance-audited and prospective. Our contributions are empirical, and we state them with their limits.

The authors are with the Department of Artificial Intelligence and Data Science, Amrita Vishwa Vidyapeetham, Coimbatore, India.
Corresponding author: [to be completed].

1) **A national-scale, leakage-resistant benchmark** of daily mandi modal-price forecasts. It covers 11,845 market–commodity–variety series of 36 commodities at four horizons, with chronological target-purged splits. It includes five naive and seasonal baselines, a per-series exponential smoother and a global gradient-boosting baseline. Results are reported with MASE, Diebold–Mariano tests and per-series win rates.
2) **The scoring function is decisive.** Between 17% and 34% of target changes are exactly zero, because mandis re-quote unchanged prices. Under absolute error, squared-loss global learners do not beat persistence at the next-report horizon, whereas an absolute-loss learner does. This is a predictable consequence of elicitation theory [7]. We quantify it at scale for this domain.
3) **A data-provenance effect.** A public mirror of AGMARKNET holds only 30% of the official reports from October 2025 onwards. On the mirror, forecasts appear to degrade and conformal intervals appear to lose calibration after a change of the publishing portal. Scored on the same (series, origin) rows, the complete official record gives near-nominal coverage, although the 7-day gain over persistence still shrinks after the change. The gap statistics are consistent with a mechanism: the mirror's median gap between reports is three times longer.
4) **Prospective evaluation.** Models trained on data up to 2023 are saved, and only then are six months of later official reports (April–October 2026) first downloaded and scored. This is a post-development hold-out, not a real-time evaluation, because the forecasts were not issued as the data arrived. We also run a retraining experiment whose final test period was not used for any retraining choice.

Gains over persistence are modest: 3–8% in MAE. We do not claim superiority over published models, whose data, horizons and protocols differ (Section IX). The prototype system (single machine) is described in the supplementary material: Spark/Kafka/HDFS pipeline, API and a six-language farmer application.

II. Related work

a) Mandi and Indian agricultural price forecasting.:

- Paul et al. compared ARIMA, GRNN, SVR, random forests and gradient boosting on daily AGMARKNET

brinjal prices from 17 Odisha markets, with Diebold–Mariano tests [1].
- A study of monthly prices for 19 commodities in 143 markets found deep recurrent models ahead of others, and included a random-walk-with-drift baseline [11].
- Hybrid, wavelet and neural approaches exist for oilseeds, pulses, onion and potato [3], [12]–[17].
- The closest study in coverage uses daily AGMARKNET data for 23 commodities in 165 markets with an 80/20 split, and explicitly decides against MASE [2].
- AGMARKNET data quality, including missing and erroneous entries, has been addressed in price-forecasting frameworks [18], [19].
- Combination methods for agricultural commodity prices have appeared in this journal [20].

b) Evaluation.: Our protocol follows established recommendations:

- scaled errors [6];
- tests of equal predictive accuracy [21];
- strong simple benchmarks [4], [5], [22];
- time-series performance estimation [23], [24];
- the catalogue of pitfalls in machine-learning forecasting [9].

The consistency of point forecasts with the scoring function [7] is particularly relevant to intermittent and zero-inflated targets [8]. Real-time data vintages are known to change forecast evaluations in macroeconomics [10]. Our provenance finding is an agricultural analogue, associated with an incomplete mirror rather than with revisions.

c) Uncertainty.: Conformal methods for time series include EnbPI [25], ensemble conformalised quantile regression [26] and adaptive conformal inference [27]. A drift-aware conformal method has recently been applied to potato prices in China [28]. We use plain split conformal with Mondrian volatility bins and *measure* its coverage.

d) Information and farmers.: Some information interventions have raised producer prices [29], [30]; others had small or no average effects [31], [32]. These results motivate cautious decision support.

e) Gap.: To our knowledge, no study of Indian mandi prices combines all four of the following:

1) national-scale daily series;
2) purged chronological evaluation against naive, seasonal and per-series statistical baselines at several horizons;
3) measured interval coverage;
4) an audit of the data source against the official record together with prospective scoring.

III. Data and provenance audit

a) Sources.: Model development used an archive of AGMARKNET daily reports for 2001-01-10 to 2026-04-21: a Kaggle dataset (khandelwalmanas/daily-commodity-prices-india, version 134, GODL-India). It has 75,984,017 rows; after quality classification and business-key de-duplication, 75,644,801 rows remain. Its columns match the official data.gov.in resource *Variety-wise Daily Market Prices Data of Commodity*. After development, an API key became available. We back-filled 7,720,662 official records for 2025-10-01 to 2026-10-22, one request series per date, 387 dates with no failure. Our second layer combines the archive before 2025-10-01 with the official record afterwards, for 82,114,083 rows in total.

b) Scope and panel.: We use 36 staple commodities: vegetables, fruits, cereals, pulses, oilseeds and cotton. They were chosen from training-period data:

1) take an a-priori list of crop-name patterns;
2) keep matches with more than 50,000 valid reports in 2021–2023;
3) drop two duplicate or processed forms;
4) keep the 36 most-reported.

An entity (series) is (state, district, market, commodity, variety). It is in the panel if it has at least 300 report days in 2021–2023, which gives 11,845 series and 21.06M feature rows. A causal sanity rule removes reports more than a factor of five from the geometric mean of the previous ten reports (28,567 reports, 0.086%).

c) Identifier drift.: Several changes in how identifiers are published would break series if left untreated:

- many series switch their grade label from *FAQ* to *Local* in January 2025. No (state, district, market, commodity, variety, date) key carries two grades in the archive since 2023, so grade is an attribute. The official record is different: there, 1.2% of entity-days (85,315) carry several grades, and a deterministic rule keeps one of them;
- markets gain an " APMC" suffix with the AGMARKNET 2.0 portal (live 2025-11-07);
- the official API writes no space before "(" and uses "Keralam" for Kerala.

One normalisation rule, shared by the batch and streaming paths, maps all of these. On 2026-04-20, all 5,152 valid archive keys are found among the 22,716 official records.

d) Mirror completeness.: Table I compares daily volumes. On the sample days in Table I the mirror holds 94–100% of official reports before October 2025 and 0–54% afterwards. By month, it holds 70.4% in October 2025 and 5.9–38.9% from November 2025 to April 2026, which is 30.0% overall for 2025-10-01 to 2026-04-21. Every mirror record in that window exists in the official data, and 99.33% have an identical modal price. The mirror is incomplete, not corrupt. Missing reports lengthen the gaps between consecutive reports of a series:

- the median gap is 3 days in the mirror and 1 day in the official record;
- 16.2% of mirror gaps exceed a week, against 2.4% in the official record.

Longer gaps push "next report" targets further ahead and thin the trailing windows the features rely on.

IV. Forecasting task and evaluation protocol

a) Task.: Entity e reports modal prices $P_{e,t}$ (INR per quintal) on irregular dates. For an origin t and $h \in \{3, 7, 14\}$ days,

TABLE I
DAILY REPORT COUNTS, OFFICIAL API VERSUS THE PUBLIC MIRROR (SAMPLE DAYS).

Date	Official	Mirror	Share
2025-03-12	22,040	21,967	0.997
2025-09-17	20,703	19,542	0.944
2025-10-15	21,899	20,611	0.941
2025-11-20	17,824	0	0.000
2025-12-17	19,566	10,559	0.540
2026-02-18	18,979	5,992	0.316
2026-04-20	22,716	5,162	0.227

the target is the first report dated in $[t+h, t+h+3]$. If there is none, there is no label. The *next* horizon is the next report, if it falls within 7 days. Models predict $y = \ln P_{e,\tau} - \ln P_{e,t}$, and errors are measured in rupees on $P_{e,t}\exp(\hat{y})$.

b) Periods (main experiment).:

- Training origins: 2016–2023.
- Validation: 2024, used for tuning, model choice and conformal calibration.
- Test origins: 2025-01-01 to 2026-04-20, scored once.

A row is kept only if its target date also lies in its period; a unit test enforces this purge. A perturbation test multiplies all future prices, and all other entities' prices, by arbitrary factors and requires every feature at the origin to be unchanged. Training uses a uniform sample of 2×10^6 rows per horizon (fixed seed).

c) Selection.: The family with the lowest validation MAE is selected, baselines included. The search space was amended twice, both times on validation evidence, before any test metric was read:

- first, after squared-loss learners tied persistence;
- second, after finding that MLlib's absolute-loss boosting is mis-scaled for log returns.

An aborted first run had already written test predictions for one horizon. They were deleted unread, and we disclose this.

d) Metrics and tests.: We report:

- MAE, RMSE, WAPE, sMAPE and bias;
- MASE, scaled by each entity's in-sample h-step naive MAE on training rows [6];
- MAE relative to persistence on identical rows;
- the share of series (at least 30 test rows) on which a model beats persistence.

Diebold–Mariano tests [21] use daily mean absolute-error differences, a Newey–West variance with lag $h + 3$ (7 for the next report) and the HLN correction. Same-day errors are aggregated first because they are correlated across markets.

e) Frozen-model and retraining evaluations.: After the official back-fill, we score the frozen models, quantiles and MASE scales on complete official data:

- on the original test period;
- on 2026-04-22 to 2026-10-22, reports first downloaded after the models were saved.

Separately, we retrain with the same search space, selection rule, sample size and seed, on a protocol fixed before the run:

- training: 2016–2024;
- validation: 2025;
- final test: 2026-01-01 to 2026-10-22, not used for any retraining choice. Its first months overlap the main test, and its later months had already been used to score the frozen models.

V. METHODS

a) Features.: Each origin carries 24 base features plus one horizon-specific seasonal feature, all computed from reports dated $\leq t$ (Table II). Missing values are set to zero. Each feature that can be missing has an indicator or count, so no fitted imputation can leak. The cross-market features use other markets' reports of the *same day*, which assumes those are published by the end of day t. We test this assumption with a variant that uses the previous day's cross-market values instead (Section VI).

b) Baselines.:

- Persistence ($\hat{y} = 0$).
- Geometric moving averages over 7 and 30 days.
- Seasonal naive: last year's level around $t+h-365$, falling back to persistence when that is missing.
- Seasonal-adjusted persistence: today's level times last year's change over the same horizon.
- Per-series simple exponential smoothing (SES) on ℓ, with the smoothing weight chosen per series on training origins.
- A global LightGBM model with an L1 objective, trained on the same features and training sample, tuned and early-stopped on validation only; predictions of three seeds are ensembled.

c) Learners.: Each horizon has its own Spark MLlib model (direct strategy). The candidates are:

- ridge and Huber regression on standardised features with one-hot commodity;
- a random forest (60 trees, depth 10);
- gradient-boosted trees with squared loss;
- gradient-boosted trees with absolute loss.

MLlib's absolute-loss boosting fits trees after the first to the sign of the residual, without a leaf line search, so each iteration moves predictions by up to the step size. We therefore train these models on the target in percent, with step sizes of 0.25–0.5, and rescale inside the persisted pipeline. A unit test shows that this recovers the conditional median of a zero-inflated target.

d) Intervals.: For the selected model, validation residuals are grouped into five bins by 30-day log-price volatility, with bin edges from validation quantiles. Each bin's 5/10/90/95% residual quantiles give 80% and 90% intervals. Exchangeability does not hold for time series, so coverage is measured on test data [27].

TABLE II
FEATURE GROUPS ($\ell = \ln P$); ALL FRAMES END AT THE ORIGIN.

Group	Features
Level and lags	ℓ_t; differences to the 1st, 2nd and 5th previous reports; days since the previous report
Rolling (7/30/90 d)	deviation from the rolling mean; standard deviation (30, 90); report counts (30, 90); distance to the 30-day max and min
Range	$(\max - \min)$/modal on day t
Seasonal	deviation from the level at $t - 365 \pm 3$; availability indicator; last year's change from $t - 365$ to $t + h - 365$
Cross-market	same-day mean 30-day deviation and last return of the commodity, nationally and within the state; number of reporting markets
Calendar	day of year (sine, cosine); day of week
Categorical	commodity

VI. POINT-FORECAST RESULTS

a) Validation: the scoring function decides.: On the 2024 validation tuning sample (600,000 rows) at the next-report horizon, persistence has an MAE of 210.7 rupees per quintal. The squared-loss learners do not beat it:

- ridge regression: 228.1;
- squared-loss boosting: 214.8;
- the random forest: 211.1, a tie with persistence;
- Huber regression: 212.5.

Absolute-loss boosting reaches 205.3. At 7 and 14 days, every learner beats persistence on validation, and the random forest is selected. This is the pattern that elicitation theory predicts when a large share of the targets is exactly zero [7]. The conditional median of the change is then often zero, while the conditional mean is not.

b) Main test.: Table III reports the selected models on 1.55–1.67 million test forecasts per horizon. Their MAE is 2.9–7.7% below persistence, and the Diebold–Mariano tests reject equal accuracy at every horizon. On test data the random forest does not beat persistence at the next report either (DM $p = 0.28$). All other baselines are worse than persistence at every horizon:

- moving averages: relative MAE 1.005–1.795;
- seasonal naive: 2.50–5.11;
- seasonal-adjusted persistence: 1.10–1.34.

The improvement is broad. The selected model beats persistence on 58.5%, 68.8%, 75.0% and 76.4% of series with at least 30 test rows.

c) Scaled errors disagree.: Averaged over rows, MASE does not favour the learned models everywhere:

- next report: 1.559 for the selected model against 1.520 for persistence;
- 7 days: 1.430 against 1.257;
- 3 days: 1.680 against 1.716;
- 14 days: 1.056 against 1.113.

The rupee gains therefore come mainly from series and days with large absolute price moves. On rows that are small relative to the series' own naive error, the models are no better than persistence and sometimes worse. We report both measures, because they answer different questions: aggregate rupee error, and an equal-weighted error per row.

d) Stronger and simpler baselines.: Two further baselines bracket the MLlib model.

- **Exponential smoothing.** Per-series simple exponential smoothing, evaluated on a random sample of 400 series, matches the global model at the next report (relative MAE 0.975 against 0.976) and at 3 days (0.962 against 0.960). The global model is clearly better only at 7 days (0.946 against 0.967) and 14 days (0.935 against 0.975).
- **LightGBM.** The global LightGBM baseline, tuned on validation only and ensembled over three seeds, is more accurate than the MLlib model at every horizon: relative MAE 0.944, 0.926, 0.915 and 0.892. The DM p against the MLlib model is at most 3×10^{-9}, and the seed-to-seed spread in MAE is at most 1.2 rupees. A single seed gives 0.946 at the next report.

The modest gains of the MLlib model are therefore partly a property of the MLlib learners and their small search space, not a ceiling for this feature set.

e) Ablation and feature timing.: Removing feature groups from the 7-day absolute-loss boosting model (not the selected random forest) has these effects:

- without the cross-market group, the relative MAE rises from 0.944 to 0.960;
- using only the three recent returns gives 0.970;
- removing the calendar, level or seasonal groups changes it by at most 0.004.

Because the cross-market features use same-day reports from other markets, we refitted the LightGBM baseline with three versions of them:

- same-day values;
- values recomputed from the previous day, which are always published before the origin;
- no cross-market features.

Table IV shows the result. Lagging costs 0.002–0.006, whereas removing the group costs 0.009–0.015. Most of the cross-market signal does not depend on same-day publication. The cross-market group accounts for about 29% of the boosting model's 7-day gain over persistence and about 17% of LightGBM's.

f) Stability across years.: In rolling-origin folds for 2021–2024 the relative MAE stays in narrow ranges: 0.970–0.976 (next), 0.952–0.960 (3 days), 0.936–0.946 (7 days) and 0.920–0.936 (14 days). Each fold retrains on all earlier years and

TABLE III
MAIN TEST, 2025-01-01 TO 2026-04-20 (MIRROR DATA). THE RELATIVE MAE IS THE MODEL'S MAE DIVIDED BY THAT OF PERSISTENCE ON IDENTICAL ROWS. DM: DIEBOLD–MARIANO TEST AGAINST PERSISTENCE.

Horizon	Persistence MAE	Selected	MAE	Rel. MAE	DM p	Series better
next	189.0	GBT, abs. loss	183.5	0.971	1×10^{-22}	58.5%
3 d	242.6	GBT, abs. loss	231.1	0.953	4×10^{-47}	68.8%
7 d	308.8	random forest	289.7	0.938	4×10^{-39}	75.0%
14 d	397.0	random forest	366.4	0.923	9×10^{-29}	76.4%

TABLE IV
RELATIVE MAE OF LIGHTGBM (SEED 1) ON THE MAIN TEST, BY TIMING OF THE CROSS-MARKET FEATURES.

Horizon	Same day	Previous day	None
next	0.946	0.948	0.955
7 d	0.918	0.922	0.932
14 d	0.893	0.899	0.908

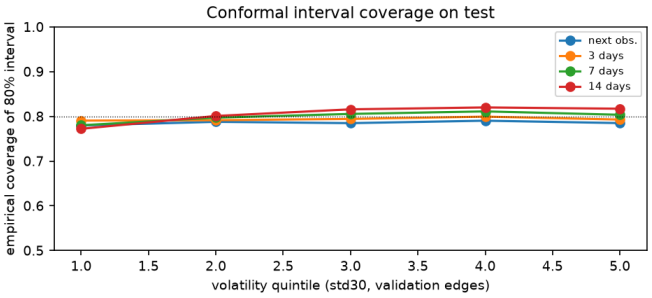


Fig. 1. Test coverage of 80% split-conformal intervals by volatility quintile (bin edges from validation).

holds out one year. The main-test result is therefore not specific to one year. Two caveats apply to these folds:

- the 2024 fold is the validation year on which the configurations were chosen, so the 2021–2023 folds also use hyper-parameters chosen on later data;
- the panel for the 2021–2023 folds was selected by its reporting density in 2021–2023. This selects series only and uses no prices.

VII. INTERVAL CALIBRATION AND THE DATA-PROVENANCE EFFECT

a) Main test.: On the main test, empirical coverage of the nominal 80% intervals is 78.6%, 79.4%, 80.0% and 80.5% for the four horizons; for the 90% intervals it is 88.9–90.5%. Across volatility quintiles, coverage stays within 77.2–82.0%. Interval widths adapt strongly, from 2–8% of the price in the calmest quintile to 45–88% in the most volatile (Figure 1).

b) Provenance effect.: Measured on the mirror after the November 2025 portal change:

- 80% coverage fell to 67.6–72.9%;
- relative MAE weakened to 0.993, 0.977, 0.968 and 0.944.

This had the hallmarks of a distribution shift. Re-scoring the *same frozen models* on the complete official record for the same period (Table V) gives:

TABLE V
FROZEN MODELS AFTER THE NOVEMBER 2025 PORTAL CHANGE (2025-11-01 TO 2026-04-20): MIRROR VERSUS COMPLETE OFFICIAL RECORD.

	next	3 d	7 d	14 d
Rel. MAE, mirror	0.993	0.977	0.968	0.944
Rel. MAE, official	0.976	0.962	0.962	0.937
80% coverage, mirror	67.6%	71.3%	72.0%	72.9%
80% coverage, official	78.0%	78.7%	78.7%	78.5%
Identical (series, origin) keys				
Rel. MAE, mirror	0.998	0.983	0.971	0.942
Rel. MAE, official	0.972	0.958	0.959	0.930
80% coverage, mirror	68.0%	71.6%	71.9%	72.3%
80% coverage, official	78.8%	79.7%	79.7%	79.3%

- coverage of 78.0–78.7%, a loss of only 1.2–2.2 points against the pre-change period;
- relative MAE of 0.976, 0.962, 0.962 and 0.937.

The two record sets differ in which rows exist: after the change the mirror has only 62,845–97,079 test rows per horizon, against 332,275–391,425 officially. We therefore also score both on the identical (series, origin) keys present in both (50,923–78,773 rows per horizon; last block of Table V). On these identical rows, mirror coverage is 68.0–72.3% and official coverage 78.8–79.7%. The calibration loss is therefore not an artefact of sample composition. Most of the apparent loss of calibration was produced by the mirror's missing reports. Accuracy is a different matter. On official data the 7-day relative MAE still rises from 0.938 before the change to 0.962 after, so the gain over persistence shrinks from 6.2% to 3.8%. Part of the accuracy shift is real. The mechanism is consistent with Section III: three-times-longer gaps make "next" targets more distant and trailing windows sparser, so realised changes exceed what validation residuals calibrated on denser series anticipate. A provenance audit is therefore a precondition for any claim about drift in this data.

VIII. PROSPECTIVE EVALUATION AND RETRAINING

a) Frozen models on later data.: The models, conformal quantiles and MASE scales of the main experiment were saved on 2026-10-22. The official back-fill that first brought reports after 2026-04-21 into the project ran on 2026-10-23. The saved models were scored, without refitting, on official reports for 2026-04-22 to 2026-10-22. This is a post-development hold-out rather than a real-time evaluation: the reports existed when the models were saved, but no development step had access to them. The scope is the same panel of series, with 0.27–0.31

million forecasts per horizon. Table VI shows the results. The relative MAE is 0.963, 0.955, 0.958 and 0.952, and the DM tests reject equal accuracy with persistence at every horizon ($p < 2 \times 10^{-15}$). The 80% intervals cover 76.9–80.1%. At 7 and 14 days the gain is smaller than in the main test (0.958 against 0.938 at 7 days), The smaller gain is not explained by model age: a model retrained up to 2024 does no better on 2026 (Table VII), so it more likely reflects conditions in 2026.

b) Does retraining help?: We retrained under the protocol fixed in Section IV: training up to 2024, validation on 2025, and a final test from 2026-01-01 to 2026-10-22 that was not used for any retraining choice. The search space, selection rule, sample size and features were the same as before. Validation selected absolute-loss boosting at the next report and at 3 days, the random forest at 7 days and squared-loss boosting at 14 days. Table VII compares the retrained and frozen models on identical final-test rows. No difference is significant (DM p between 0.12 and 0.99). The retrained model is numerically worse at 14 days (0.950 against 0.946). Both models beat persistence at every horizon ($p \leq 1.4 \times 10^{-12}$). The moving-average, seasonal-naive and seasonal-adjusted baselines are again worse than persistence (relative MAE 1.007–3.98).

c) Adopting LightGBM.: Before training, we committed a rule for replacing the served model. The candidate had to beat the frozen model at 7 days after a Holm correction, be significantly worse at no horizon, and keep 80% coverage within 75–85%, losing no more than two points relative to the frozen model. Its published forecasts also had to reproduce its offline predictions exactly. A LightGBM candidate trained under the retraining protocol (validation used only for the number of leaves, early stopping and calibration) reached relative MAE 0.944, 0.933, 0.934 and 0.922 on the rows of Table VII. It was better than both Spark models at every horizon (Holm-adjusted $p \leq 4 \times 10^{-16}$), with 80% coverage of 77.5–79.0%. It was adopted. Because this final period had already been used twice, and an earlier LightGBM result on overlapping data had motivated the test, the adoption is not an independent confirmation.

d) Interpretation and disclosure.: Two more years of data did not measurably improve this model class on 2026. This is a single refit on one final period. It does not show that retraining never helps, and it does not make the frozen models current: they encode no information after 2023. The final test is not pristine. January to April 2026 belonged to the main test, and April to October 2026 had already been used to score the frozen models (above), before the retraining protocol was fixed. No choice in the retraining run used any 2026 data.

IX. Decision-rule back-test, discussion and limitations

a) Decision rule.: The application maps the 7-day 80% interval $[\ell_7, u_7]$ (on the log-return scale) to cautious market-outlook cards. Holding cost is $c = 1\% + 5\%$/week for perishables (1%/week for durables); these costs are placeholders that the farmer can edit.

- HOLD if $\ell_7 \geq \ln(1 + c)$;
- SELL if $u_7 \leq 0$;
- COMPARE if a market in the same state reported a price at least 5% higher within three days, before transport costs (which the farmer enters; the default is zero);
- WATCH otherwise.

On the main test the rule (configuration version 2026-10-22.1) abstains on 98.5% of report-days. HOLD fires on 0.5%, and the realised price exceeds today's plus cost 80.0% of the time. SELL fires on 1.0%, with a hit rate of 67.8%. HOLD fires mostly after reports far below a series' recent level that then revert (median realised change $+46.8\%$). Some of these may be reporting errors below the sanity threshold. These are hit rates on published modal prices, not farm-gate profits, and COMPARE is not back-tested because transport costs are unknown.

b) Comparison with published results.: Published mandi studies use different markets, horizons, aggregation and splits. Their MAPE values (e.g. [2]) are not comparable with our WAPE. The comparisons we can defend are internal:

- learned models beat persistence by 3–8%;
- they match a per-series smoother at 1–3 days;
- cross-market features account for 17–29% of the 7-day gain, depending on the learner;
- a global LightGBM beats the MLlib model by a further 0.023–0.031 in relative MAE.

These effects are far smaller than gains over ARIMA reported elsewhere, which suggests that baselines and protocol may explain much of the spread in reported accuracy.

c) Threats to validity.:

- Model choice, conformal calibration and the panel definition use pre-test data, but the conformal quantiles are calibrated on the same validation year used for selection.
- For backtest folds within 2021–2023, the panel was selected on reporting density in those years. No prices are involved.
- Diebold–Mariano tests treat daily cross-sectional mean losses as one series. We report one test per model and horizon against persistence without a multiplicity correction; the conclusions are unchanged with a Newey–West lag of 20 ($p \leq 8 \times 10^{-13}$ at every horizon).
- Results describe well-reported mandis of 36 commodities.
- Modal price is the published summary, not an individual farmer's price.

d) Limitations.:

- The MLlib search space is small (8 configurations per horizon), and the LightGBM model now in service was adopted on a final period that had already been used twice (Section VIII).
- Only one retraining was evaluated (Section VIII).
- No deep or foundation-model baselines were run.
- Arrivals and weather covariates are absent.
- Adaptive conformal methods were not implemented.
- A validated refresh job exists, but no schedule is installed. Freshness therefore depends on operator runs.

TABLE VI
FROZEN MODELS (TRAINED UP TO 2023), WITH RELATIVE MAE AND 80% COVERAGE. THE FIRST BLOCK RE-SCORES THE MAIN TEST PERIOD ON THE COMPLETE OFFICIAL RECORD. THE SECOND IS PROSPECTIVE (2026-04-22 TO 2026-10-22).

	next	3 d	7 d	14 d
Rel. MAE, main test (official)	0.972	0.954	0.943	0.925
80% coverage, main test (official)	79.2%	79.6%	80.0%	80.3%
Rel. MAE, prospective	0.963	0.955	0.958	0.952
80% coverage, prospective	76.9%	78.8%	79.6%	80.1%
Series better than persistence, prospective[a]	59.3%	64.1%	66.6%	66.0%

[a] Among 2,497–2,701 series with at least 30 prospective rows (about 8,600 in the main test).

TABLE VII
FINAL TEST, 2026-01-01 TO 2026-10-22, WITH IDENTICAL ROWS FOR BOTH MODELS. THE DM TEST COMPARES THE RETRAINED MODEL WITH THE FROZEN ONE.

Horizon	Test rows	Rel. MAE, frozen	Rel. MAE, retrained	DM p	80% coverage (frozen / retrained)
next	553,801	0.965	0.966	0.12	77.0% / 77.5%
3 d	531,321	0.955	0.955	0.99	78.6% / 78.8%
7 d	519,251	0.958	0.958	0.74	79.3% / 78.9%
14 d	496,306	0.946	0.950	0.35	79.7% / 78.1%

e) Ethics.: Forecasts can hurt farmers who over-trust them. The interface therefore:

- shows intervals, measured error and data age;
- defaults to WATCH;
- never states a guaranteed price.

No personal data is collected. Interface translations into five Indian languages are machine-drafted and await native-speaker review. Welfare effects are not claimed [31], [32].

X. CONCLUSIONS

On a national panel of Indian mandi series, evaluated without leakage and against honest baselines:

- learned models improve on persistence by 3–8% in MAE;
- they improve on a per-series smoother only beyond three days;
- cross-market features contribute 17–29% of the 7-day gain over persistence, and most of that survives a one-day lag;
- a global LightGBM is more accurate than the MLlib models, so their gains are not a ceiling.

The training loss must match the scoring rule when many target changes are exactly zero. An incomplete public mirror can manufacture an apparent loss of calibration that disappears on identical rows of the official record, while a smaller accuracy shift remains. Frozen models keep a significant, if smaller, advantage on later official reports that were absent from the development data. One refit on two further years gave no measurable gain on a final period not used for any retraining choice. Future work, in order of expected value:

1) rolling or scheduled retraining evaluated with rolling origins;
2) adaptive conformal intervals;
3) arrivals and weather covariates;
4) a field study of the decision cards.

a) Data and code availability.: All code, configuration, tests, experiment manifests and result files are in the project repository. The repository is public at https://github.com/kaushalrog/AG-DATA, and the release commit will be tagged at acceptance. A static demonstration of the farmer application, serving a bundled snapshot of real reports, is at https://kaushalrog.github.io/AG-DATA/. The experiment manifests of the main run record an uncommitted working tree (`git_dirty`) and a parent commit that predates the forecasting code; the forecasting code was committed afterwards, so the exact code state of that run cannot be proven from the manifest alone. The data sources are:

- the public archive (Kaggle dataset above);
- the official data.gov.in resource `35985678-0d79-46b4-9ed6-6f13308a1d24`, which requires a free key.

Every number in the tables is read from result files under `results/`. These files are written by the Spark jobs and by the scripts in `py/agri/analysis`. The mapping from each number to its file is in `docs/VERIFIED_METRICS.md`.

ACKNOWLEDGMENT

The authors thank Dr. Sreeja (Amrita Vishwa Vidyapeetham) for guidance. AGMARKNET data are published by the Government of India under the GODL-India licence.

REFERENCES

[1] R. K. Paul, M. Yeasin, P. Kumar, P. Kumar, M. Balasubramanian, H. S. Roy, A. K. Paul, and A. Gupta, "Machine learning techniques for forecasting agricultural prices: A case of brinjal in Odisha, India," *PLOS ONE*, vol. 17, no. 7, p. e0270553, 2022.

[2] R. L. Manogna, V. Dharmaji, and S. Sarang, "Enhancing agricultural commodity price forecasting with deep learning," *Scientific Reports*, vol. 15, no. 1, p. 20903, 2025.

[3] P. Kumar, P. S. Badal, R. K. Paul, G. K. Jha, P. Venkatesh, Kamalvanshi, P. Anbukani, M. Balasubramanian, and P. Patel, "Forecasting onion price for Varanasi market of Uttar Pradesh, India," *The Indian Journal of Agricultural Sciences*, vol. 91, no. 2, pp. 249–253, 2021.

[4] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and machine learning forecasting methods: Concerns and ways forward," *PLOS ONE*, vol. 13, no. 3, p. e0194889, 2018.
[5] ——, "The M4 competition: 100,000 time series and 61 forecasting methods," *International Journal of Forecasting*, vol. 36, no. 1, pp. 54–74, 2020.
[6] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.
[7] T. Gneiting, "Making and evaluating point forecasts," *Journal of the American Statistical Association*, vol. 106, no. 494, pp. 746–762, 2011.
[8] S. Kolassa, "Evaluating predictive count data distributions in retail sales forecasting," *International Journal of Forecasting*, vol. 32, no. 3, pp. 788–803, 2016.
[9] H. Hewamalage, K. Ackermann, and C. Bergmeir, "Forecast evaluation for data scientists: common pitfalls and best practices," *Data Mining and Knowledge Discovery*, vol. 37, no. 2, pp. 788–832, 2022, published online 2022; appears in volume 37 (2023).
[10] K. O. Bokun, L. E. Jackson, K. L. Kliesen, and M. T. Owyang, "Fred-sd: A real-time database for state-level data with forecasting applications," *International Journal of Forecasting*, vol. 39, no. 1, pp. 279–297, 2023.
[11] R. K. Paul, M. Yeasin, C. Tamilselvi, A. K. Paul, P. Sharma, and P. S. Birthal, "Can deep learning models enhance the accuracy of agricultural price forecasting? Insights from India," *Intelligent Systems in Accounting, Finance and Management*, vol. 32, no. 1, p. e70002, 2025.
[12] R. K. Paul and S. Garai, "Performance comparison of wavelets-based machine learning technique for forecasting agricultural commodity prices," *Soft Computing*, vol. 25, no. 20, pp. 12 857–12 873, 2021.
[13] G. K. Jha and K. Sinha, "Time-delay neural networks for time series prediction: an application to the monthly wholesale price of oilseeds in India," *Neural Computing and Applications*, vol. 24, no. 3-4, pp. 563–571, 2012, published online 2012; appears in volume 24 (issue 3–4).
[14] ——, "Agricultural price forecasting using neural network model: An innovative information delivery system," *Agricultural Economics Research Review*, vol. 26, no. 2, pp. 229–239, 2013. [Online]. Available: http://ageconsearch.umn.edu/record/162150
[15] P. Anjoy, R. K. Paul, K. Sinha, A. K. Paul, and M. Ray, "A hybrid wavelet based neural networks model for predicting monthly WPI of pulses in India," *The Indian Journal of Agricultural Sciences*, vol. 87, no. 6, pp. 834–839, 2017.
[16] A. Praveenkumar, G. K. Jha, S. D. Madival, A. Lama, and R. R. Kumar, "Deep learning approaches for potato price forecasting: Comparative analysis of LSTM, Bi-LSTM, and AM-LSTM models," *Potato Research*, vol. 68, no. 2, pp. 1941–1963, 2024, published online 2024; appears in volume 68.
[17] R. Jaiswal, G. K. Jha, R. R. Kumar, and K. Choudhary, "Deep long short-term memory based model for agricultural price forecasting," *Neural Computing and Applications*, vol. 34, no. 6, pp. 4661–4676, 2021, published online 2021; appears in volume 34.
[18] A. Jain, S. Marvaniya, S. Godbole, and V. Munigala, "A framework for crop price forecasting in emerging economies by analyzing the quality of time-series data," arXiv:2009.04171, 2020.
[19] H. Guo, A. Woodruff, and A. Yadav, "Improving lives of indebted farmers using deep learning: Predicting agricultural produce prices using convolutional neural networks," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 08, pp. 13 294–13 299, 2020.
[20] J. Wang, Z. Wang, X. Li, and H. Zhou, "Artificial bee colony-based combination approach to forecasting agricultural commodity prices," *International Journal of Forecasting*, vol. 38, no. 1, pp. 21–34, 2022.
[21] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
[22] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.
[23] C. Bergmeir, R. J. Hyndman, and B. Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction," *Computational Statistics & Data Analysis*, vol. 120, pp. 70–83, 2018.
[24] V. Cerqueira, L. Torgo, and I. Mozetič, "Evaluating time series forecasting models: an empirical study on performance estimation methods," *Machine Learning*, vol. 109, no. 11, pp. 1997–2028, 2020.
[25] C. Xu and Y. Xie, "Conformal prediction for time series," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 10, pp. 11 575–11 587, 2023.
[26] V. Jensen, F. M. Bianchi, and S. N. Anfinsen, "Ensemble conformalized quantile regression for probabilistic time series forecasting," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 7, pp. 9014–9025, 2024.
[27] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021, preprint: arXiv:2106.00170, doi:10.48550/arXiv.2106.00170. [Online]. Available: https://proceedings.neurips.cc/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html
[28] J. Ouyang and C. Tang, "Drift-aware conformal prediction for reliable potato price forecasting under nonstationary markets," *Potato Research*, vol. 69, no. 3, 2026.
[29] R. Jensen, "The digital provide: Information (technology), market performance, and welfare in the South Indian fisheries sector," *The Quarterly Journal of Economics*, vol. 122, no. 3, pp. 879–924, 2007.
[30] A. Goyal, "Information, direct access to farmers, and rural market performance in Central India," *American Economic Journal: Applied Economics*, vol. 2, no. 3, pp. 22–45, 2010.
[31] M. Fafchamps and B. Minten, "Impact of SMS-based agricultural information on Indian farmers," *The World Bank Economic Review*, vol. 26, no. 3, pp. 383–414, 2012.
[32] S. Mitra, D. Mookherjee, M. Torero, and S. Visaria, "Asymmetric information and middleman margins: An experiment with Indian potato farmers," *The Review of Economics and Statistics*, vol. 100, no. 1, pp. 1–13, 2018.